1.13 Algebra: Algebraic Fractions

Chapter 7 · Original slide 4 · 1: Simplifying Algebraic Fractions

Recall that you can simplify fractions by dividing the numerator and denominator by a common factor.

$\frac{x^2-1}{x^2+x} =$

Note: Do not leave 1 in the denominator!

$\frac{2x^2+11x+12}{x^2+9x+20} =$

Tip: $\frac{a-b}{b-a} = -1$

Hint: To identify common factors we need to factorise first.

$\frac{x^2+3x+2}{x+1} =$

Tip: Factorise the easier one first because it provides clues to the factorisation of the other.

$\frac{4-x^2}{x^2+2x-8} =$

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2 Simplify these fractions as far as possible:

m $\frac{2x^2+3x+1}{x^2-x-2}$ **n** $\frac{x^2+6x+8}{3x^2+7x+2}$ **o** $\frac{2x^2-5x-3}{2x^2-9x+9}$

3 $\frac{6x^3+3x^2-84x}{6x^2-33x+42} = \frac{ax(x+b)}{x+c}$, where a, b and c are constants.
Work out the values of a, b and c.